Ian Sequeira

Santa Barbara, CA | ian_sequeira@ucsb.edu | ian-sequ7.github.io/ian_sequeira |
linkedin.com/in/ian-sequ7 | github.com/ian-sequ7

Education

University of California, Santa Barbara — Santa Barbara, CA
B.S. Statistics & Data Science, B.A. Economics — *Sept 2023 – June 2027*

- GPA: 3.82 / 4.0 | Dean's Honors (6 quarters)
- Coursework: Machine Learning, Design of Experiments, Stochastic Processes, Regression Analysis, Econometrics, Probability & Statistics, Proof Writing

Research Experience

UCSB Department of Statistics & Applied Probability — Santa Barbara, CA
Undergraduate Researcher – Prof. Annie Qu & Prof. Xiaowu Dai — *Apr 2026 – Present*

- Developing **Compositional Coordination Confidence** — anytime-valid confidence guarantees for multi-agent LLM pipelines via typed e-processes that compose across DAG edges
- Proving composition theorems; Monte Carlo simulations validated the framework across sequential, fan-in, and fan-out agent topologies

UC Irvine Summer Institute for Biostatistics — Irvine, CA
Research Fellow – Prof. Dan Gillen & Dr. Josh Grill — *June 2025 – Aug 2025*

- Constructed GLM, ridge, and CART classifiers in R identifying Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 trial participants (0.835 AUC)
- Developed sample size reduction framework achieving 80% power with 188 vs. 500+ participants
- Co-author on forthcoming biomarker-guided trial design paper in *Journal of Prevention of Alzheimer's Disease (JPAD)*, 2026; research presented at *CTAD 2025*, San Diego

Work Experience

Alida Inc. — Remote
Student Intern Data Engineer — *Sept 2024 – Present*

- Built multi-mode AI data exploration platform over 47 healthcare datasets using multi-agent routing, graph analysis, and automated SQL generation in Databricks
- Extended platform with spec-driven ML prediction pipeline featuring typed contracts, dual-layer verification, and fault-tolerant LLM orchestration with AutoML training

Projects

Healthcare Financial Toxicity Prediction — Mar 2026
R, XGBoost, BART, SHAP, fairmodels

- PSTAT 231 (graduate Statistical ML) final project: predicted catastrophic out-of-pocket burden on 20,962 MEPS respondents; benchmarked 6 classifiers under 10-fold CV (XGBoost ROC-AUC 0.874)
- Audited subgroup fairness with DALEX and produced SHAP-based odds-ratio interpretations

Research Advisor Finder — Nov 2025 – Feb 2026
FastAPI, PostgreSQL, pgvector, Next.js, OpenAI, Claude API

- Built full-stack faculty discovery platform matching student research interests to 15k+ professors across 12 STEM fields via hybrid semantic search and CV-driven interest extraction

Leadership and Activities

Project Executive, ACM Industry, UCSB — Sept 2025 – Present

- Leading cross-functional student teams delivering applied data science and software solutions for industry partner Turing

Finance Committee, CampMed, UCSB — Sept 2025 – Present

- Coordinating fundraising and event logistics for health-equity outreach to underrepresented high school students

Technical Skills

Programming: R, Python, SQL, C++, JavaScript/TypeScript, HTML & CSS, LaTeX
Frameworks & Tools: tidymodels, xgboost, rstan, scikit-learn, PyTorch, SHAP, Databricks, LangGraph
Languages: English, Spanish | **Memberships**: ASA, IMS, UCSB Innovators Circle